

Compliance Query Triage Agent

AI Product Mini-PRD | Prepared by Sharmila Auti | June 2026

1. Problem Statement

Front-line banking support agents and compliance reviewers currently handle routine policy queries manually — answering questions about account opening, KYC verification, card services, fees, and account management individually. This creates inconsistent response quality dependent on individual agent knowledge, slow turnaround averaging 10-15 minutes per query, and compliance risk from ungrounded or outdated policy references. A governed AI agent that classifies queries, retrieves the relevant policy snippet, and routes only uncertain or high-risk cases to human reviewers can reduce handling time to under 10 seconds while maintaining full auditability and source traceability.

2. Why Agentic, Not a Simple Chatbot or FAQ Search

A scripted FAQ bot can answer fixed questions with fixed answers. This problem needs more than retrieval — it needs judgement at each step:

- **Classification:** The query type and risk category are not always obvious from keywords alone, so the agent reasons about intent before acting.
- **Grounded drafting:** The response must be traceable to an approved policy source, not generated from general knowledge — this is what makes it usable in a regulated setting.
- **Confidence and risk scoring:** A second LLM pass (LLM-as-judge) assesses the agent's own certainty and the sensitivity of the topic, rather than treating every query the same way.
- **Conditional autonomy:** Only low-risk, high-confidence cases grounded in policy are auto-resolved. Everything else routes to a human — this is the agentic part: the workflow changes its own path based on what it finds at runtime.

This pattern — multi-step LLM reasoning, retrieval-augmented generation (RAG), self-scoring, and human-in-the-loop routing — is what distinguishes an agentic workflow from a simple automation.

3. MVP Scope

In Scope

- **Single input channel:** browser-based form trigger (simulates a customer or agent submitting a query)
- **Fixed policy knowledge base:** 20-entry Customer Banking FAQ Policy Document (Google Sheets), covering account opening, KYC, card services, payments, fees, and account management

- Binary routing: auto-resolve (high confidence, low risk, FAQ-grounded) or human review queue (all other cases)
- Dual Google Sheets output: Pending Review tab for human reviewer, Audit Log tab for every run
- LLM: GPT-4o-mini via OpenAI API, orchestrated in n8n

Explicitly Out of Scope (Future Releases)

- Multi-channel intake (email, chat widget, CRM webhook)
- Vector database / live policy document sync — current version embeds FAQ as text in the LLM prompt
- Multi-tier escalation (e.g. Level 1 reviewer to Level 2 compliance officer)
- Multi-language support
- Analytics dashboard beyond the raw audit log

Known Limitations

- Static knowledge base: The FAQ policy document is loaded as plain text into the LLM prompt on every run. Any policy update requires manually editing the Google Sheet — there is no live sync or versioning.
- Prompt-based RAG only: This is not a true vector search implementation. The full FAQ is passed as context each time, which limits scalability beyond approximately 30-40 Q&A pairs before hitting LLM context constraints.
- LLM non-determinism: The confidence score and risk flag can vary slightly between runs for the same query, making the threshold a probabilistic gate rather than a fully deterministic rule.
- No feedback loop: Reviewer edits and rejections are captured in the sheet but not yet used to improve the model or prompt over time.

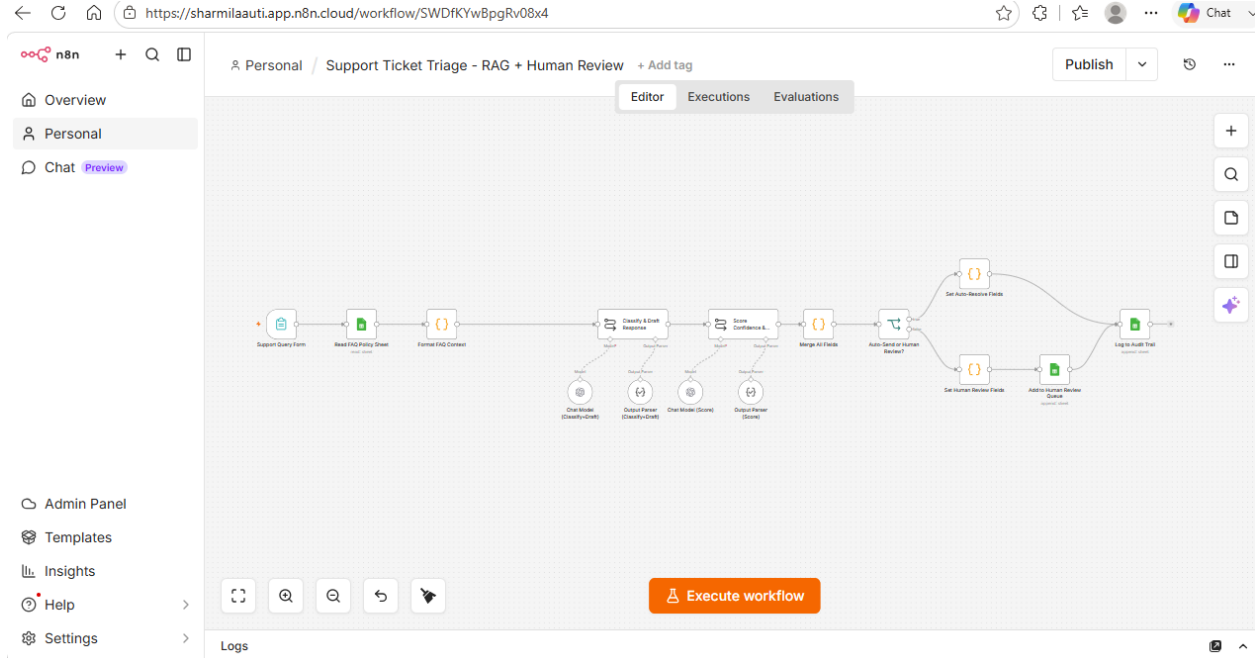
MVP principle applied: prove classification accuracy, grounding reliability, and governance acceptability on a narrow slice before expanding scope.

4. Workflow Design (Built in n8n)

Technology Stack

n8n (workflow orchestration) | GPT-4o-mini via OpenAI API (LLM reasoning and scoring) | Google Sheets (policy knowledge base, Pending Review queue, Audit Log) | n8n Form Trigger (query intake)

Eight-Node Agentic Pipeline



- Step 1 — Trigger: Form trigger captures the user query with timestamp and unique requestId.
- Step 2 — Read FAQ: Google Sheets node reads all rows from the Customer Banking FAQ Policy Document (20 Q&A pairs), filtered to remove blank rows.
- Step 3 — Format FAQ Context: Code node aggregates all FAQ rows into a single structured text block passed to the LLM as policy context.
- Step 4 — Classify & Draft Response: LLM Chain (GPT-4o-mini) classifies the query type (payments / account_access / kyc / card_services / fees / complaint / other), drafts a policy-grounded reply, and identifies the exact FAQ source used. Semantic matching is used — partial or rephrased queries are matched by intent, not exact wording.
- Step 5 — Score Confidence & Risk: A second LLM-as-judge call rates confidence 1-5 (based on whether the answer is grounded in the policy document) and flags risk level (low / medium / high) based on whether the query touches fraud, KYC, complaints, or legal topics.
- Step 6 — Merge All Fields: Code node assembles all fields from upstream nodes into one flat item for downstream use.
- Step 7 — Decision Gate: IF node evaluates three conditions (AND): confidence ≥ 3 , risk_flag = low, source_used $\neq$ none. All three must pass for auto-resolve; any failure routes to human review.
- Step 8a — Auto-Resolve: Adds branch=auto_resolved and finalStatus=resolved. Logs to Audit Trail.
- Step 8b — Human Review: Adds branch=human_review and finalStatus=pending_review. Appends full record (including draft response and AI reasoning) to Pending Review sheet, then logs to Audit Trail.

5. Governance and Risk Controls

Human Oversight

Every query that does not meet all three auto-resolve conditions is written to a Pending Review Google Sheet. The record includes the original query, AI-drafted response, source FAQ entry, confidence score, risk flag, and reasoning. The reviewer fills in reviewer_action (approve / edit / reject) and reviewer_notes before the response is sent.

Source Traceability

Every drafted response logs source_used — the exact question text from the FAQ policy document that grounded the answer. If no FAQ entry matched, source_used is set to 'none' and the query is automatically escalated regardless of confidence score.

Audit Trail

Every execution is logged to the Audit Log sheet with: requestId, receivedAt, query, category, draft_response, source_used, confidence_score, risk_flag, risk_category, reasoning, branch, finalStatus. This provides a complete, reviewable record of every AI decision.

Risk-Based Escalation Rules

- Confidence score below 3 (out of 5) — mandatory human review
- Risk flag = medium or high (fraud, KYC, complaints, legal) — mandatory human review
- source_used = 'none' (no FAQ match found) — mandatory human review regardless of confidence score

Model Risk / Explainability

This is not a black-box decision. Every output is reviewable, scoped to a single known knowledge domain (the FAQ document), and gated by a deterministic confidence threshold. The LLM does not have access to live customer data.

6. Success Metrics (KPIs)

Execution time figures are based on 12 test runs completed in June 2026 using the n8n workflow with GPT-4o-mini. All other targets are forward-looking estimates based on demo results.

Metric	Baseline (Manual)	Target (With Agent)	How Measured
Average handling time per query	10-15 minutes	< 10 seconds (~9 seconds average across 12 test runs)	n8n Executions tab — duration shown per run after each execution
Auto-resolution rate	0% (fully manual)	60-70% of routine FAQ queries	Count of branch=auto_resolve

			d vs total entries in Audit Log
Escalation accuracy	n/a	> 85% correctly flagged	Manual review of sampled escalations in Pending Review sheet
Response accuracy (FAQ-grounded queries)	Varies by agent knowledge	> 90% accepted without edit by reviewer	reviewer_action = 'approve' rate in Pending Review sheet

7. Rollout Plan

Phase 1 — Pilot: One query category (account management), internal team only, manual fallback always available. Weekly review of escalation log to calibrate confidence threshold and identify prompt improvements.

Phase 2 — Controlled Expansion: Add two to three further query categories (payments, KYC). Tighten or relax confidence threshold based on Phase 1 accuracy data. Introduce reviewer SLA tracking.

Phase 3 — Scale: Broaden to all FAQ categories. Replace static FAQ text with a vector store for live policy updates. Formalise governance sign-off with Risk and Compliance. Hand off audit log to a permanent reporting view.

Change management note: Front-line staff and compliance reviewers need to trust the escalation logic before adoption sticks. A transparent, auditable Pending Review queue — where they can see exactly why the AI flagged a query and what it drafted — is the primary trust-building mechanism.

8. What I Would Do Differently at Scale

- Replace the static FAQ text in the LLM prompt with a proper vector store (e.g. Pinecone or pgvector) so the knowledge base can be updated without redeploying the workflow — critical for a live policy environment.
- Add a feedback loop where reviewer edits and rejections are captured and used to refine prompt examples or retrain a fine-tuned classifier over time.
- Introduce a confidence-threshold tuning process owned jointly by product and risk, reviewed monthly against escalation accuracy data — the threshold should be a governed parameter, not a hardcoded number.
- Add monitoring and alerting for classification drift — a spike in escalation rate could signal a policy change upstream that the agent has not been updated to reflect.
- Separate the scoring LLM call into a dedicated risk-assessment microservice so it can be audited, versioned, and updated independently of the drafting step.

